A Single-Mode Occupation-Number Model for Temperature-Dependent Zero-Field Splitting in NV Centers

Reddouane BERRAMDANE[1]

[1]Independent Researcher

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Abstract

The temperature dependence of the zero-field splitting $D(T)$ in nitrogen-vacancy (NV) centers in diamond is central to quantum sensing, yet it is often described by empirical polynomial fits with limited physical interpretability. We present a compact effective model in which $D(T)$ is governed by the Bose–Einstein occupation of a single representative phonon mode. The model contains three parameters: the zero-temperature limit D_0, a coupling amplitude A, and an effective phonon frequency ω (expressed in cm^{-1}).

Fitted to the high-precision dataset of Ouyang et al. in the range 298–383 K, the model matches the best published second-order polynomial within the reported experimental uncertainties of that dataset while retaining direct physical meaning. The fitted frequency, $\omega = 711 \pm 48\,\mathrm{cm}^{-1}$, lies within the physically plausible phonon range of diamond. Calibrated only in the 298–383 K window, the model reproduces the parametrization of Liu et al. up to 1000 K with $R^2 = 0.998185$ and a maximum deviation below 2 MHz, whereas a power-law calibration performs poorly outside the fitting window.

These results establish the model as a compact, physically interpretable alternative to purely empirical descriptions of $D(T)$ and suggest that the same phonon-occupation framework may be extended to spin-lattice relaxation $1/T_1(T)$.

1 Introduction

The nitrogen-vacancy (NV) center in diamond is one of the most versatile solid-state quantum sensors [1, 2]. The temperature dependence of the ground-state zero-field splitting, $D(T)$, underpins NV-based nanothermometry [3–5] and must be accounted for in precision measurements under varying thermal conditions. Although $D(T)$ has been measured over a wide range of temperatures [6–10], the most common descriptions remain empirical polynomial fits.

A physical interpretation links the temperature shift to lattice expansion and electron–phonon coupling [11]. Recent work by Cambria et al. [12] showed that two effective phonon modes spanning the acoustic and optical branches of the diamond spectrum are sufficient to describe $D(T)$ quantitatively. Building on this idea, we show that a single effective mode can achieve comparable accuracy within the calibration window and strong extrapolation performance without increasing parameter complexity. This parsimony is attractive because it yields a model with direct physical meaning and a controlled high-temperature asymptote.

2 Model

We model the temperature-dependent zero-field splitting as

$$D(T) = D_0 + A\,n(\omega, T), \tag{1}$$

where

$$n(\omega, T) = \frac{1}{\exp\left(\frac{\hbar\omega}{k_B T}\right) - 1} \tag{2}$$

is the Bose–Einstein occupation number. Here D_0 is the zero-temperature extrapolation, A is a coupling amplitude, and ω is an effective phonon frequency.

In this manuscript, ω is expressed in cm^{-1}; accordingly, the exponential argument $\hbar\omega/k_B T$ is implemented in the equivalent wavenumber form $\omega/(k_{B,\mathrm{cm}}\,T)$, where $k_{B,\mathrm{cm}} = 0.69503476\ \mathrm{cm}^{-1}\,\mathrm{K}^{-1}$ is the Boltzmann constant in wavenumber units ($k_{B,\mathrm{cm}} = k_B/hc$, with h Planck's constant and c the speed of light). The two forms are exactly equivalent.

At high temperature ($k_B T \gg \hbar\omega$), $D(T)$ becomes approximately linear in T. At low temperature, $D(T)$ approaches D_0 exponentially. The single-mode ansatz is adopted here on grounds of parsimony: it achieves strong fit quality within the calibration window while preserving a physically interpretable structure.

3 Results

3.1 Calibration on high-precision data

We fit the model to the dataset of Ouyang et al. [10], taken from Table 1 of that work (18 points, 298.15–383.15 K, uncertainties $u(D) = (3.6\text{–}8.5) \times 10^{-5}$ GHz). The fitted parameters are

$$D_0 = 2.8764\,\mathrm{GHz}, \tag{3}$$
$$A = -0.1889 \pm 0.0299\,\mathrm{GHz}, \tag{4}$$
$$\omega = 711 \pm 48\,\mathrm{cm}^{-1}. \tag{5}$$

The corresponding fit quality is $R^2 = 0.999946$ with RMSE 1.75×10^{-5} GHz.

The Gaussian approximation to the parameter posterior is justified by the small relative uncertainties: $\sigma(D_0)/D_0 < 0.02\%$, $\sigma(A)/|A| \approx 16\%$, and $\sigma(\omega)/\omega \approx 7\%$.

The extracted D_0 is slightly lower than the commonly cited $D(0) \approx 2.8775$–2.8780 GHz [11]. The difference of approximately 1–2 MHz lies within the extrapolation uncertainty.

All three models achieve comparable fit quality, and all reported RMSE values are below the reported measurement uncertainty. The main distinction is interpretability: the present model uses physically meaningful parameters, whereas the polynomial coefficients are purely empirical.

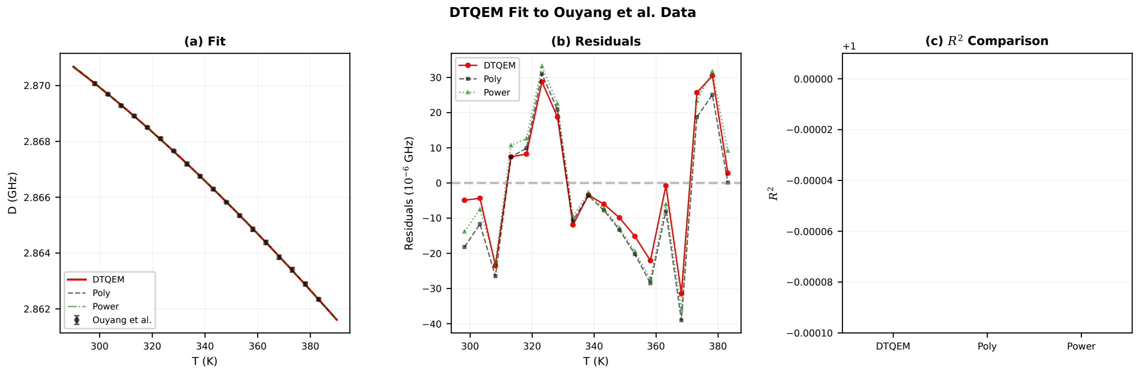

Figure 1: Results of fitting three models to the Ouyang et al. dataset (18 points, 298–383 K). (a) Model curves overlaid on the experimental data with error bars representing the reported standard uncertainties $u(D)$. (b) Fit residuals ($D_{\text{data}} - D_{\text{model}}$) expressed in units of 10^{-6} GHz; all three models produce residuals well within the measurement uncertainty. (c) R^2 values for the three models; the y-axis is truncated near unity to resolve small differences. All RMSE values are below the reported measurement uncertainty.

Table 1: Model comparison in the 298–383 K calibration window.

Model	R^2	RMSE (GHz)
This work	0.999946	1.75×10^{-5}
Ouyang polynomial	0.999932	1.95×10^{-5}
Power law	0.999930	1.98×10^{-5}

Parameter estimation and uncertainty propagation

Model parameters were estimated by weighted nonlinear least squares using `scipy.optimize.curve_fit` with the reported standard uncertainties $u(D)$ of Ouyang et al. as 1σ weights (`absolute_sigma=True`), yielding the parameter covariance matrix Σ directly in physical units. The 1σ parameter uncertainties are obtained as $\sigma_i = \sqrt{\Sigma_{ii}}$.

To propagate parameter uncertainty to the model predictions, we drew $N = 10\,000$ independent samples from the joint Gaussian posterior $\mathcal{N}(\hat{\boldsymbol{p}}, \Sigma)$, evaluated $D(T)$ for each sample over the full extrapolation range 280–1020 K, and computed pointwise 2.5th and 97.5th percentiles. The resulting shaded regions in Fig. 2(a) represent *95% parameter-uncertainty bands*; they reflect only the propagated uncertainty in the fitted parameters and do not include measurement noise at the extrapolation temperatures. The same procedure was applied independently to the power-law model to enable a direct visual comparison. The Liu et al. values used for residual evaluation were digitized from Fig. 2(c) of Ref. [14] and carry no independently assigned measurement uncertainty; the ± 5 MHz shading in Fig. 2(b) is a visual guide only.

3.2 Predictive validation: extrapolation to 1000 K

We apply the parameters obtained from the Ouyang calibration directly to the high-temperature $D(T)$ parametrization of Liu et al. [14]. The values were sampled from the published curve using digital extraction and are used here as benchmark data.

The deviation remains below 2 MHz across the full range. In contrast, a power-law

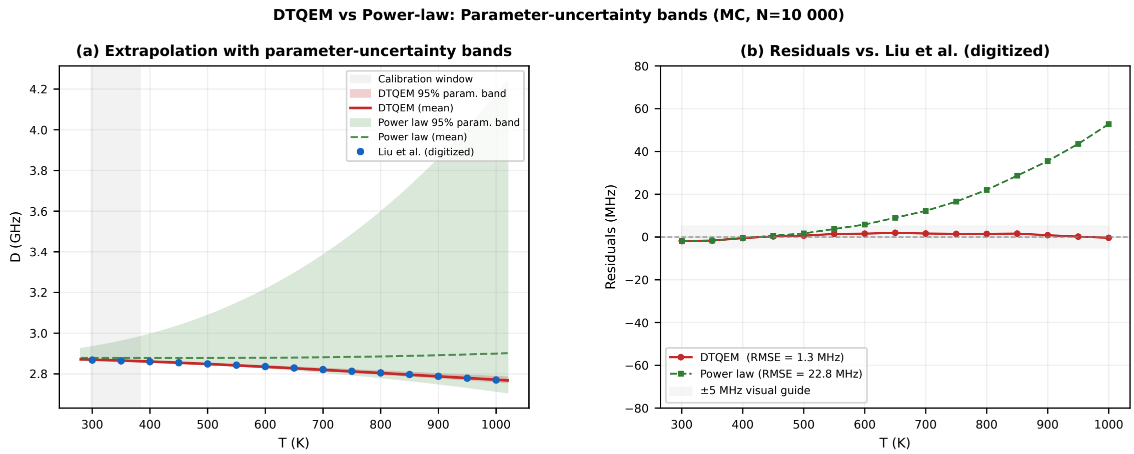

Figure 2: Extrapolation of the DTQEM and power-law models from the 298–383 K calibration window (gray shading) to the Liu et al. parametrization up to 1000 K. (a) Model predictions with 95% parameter-uncertainty bands obtained by Monte Carlo sampling ($N = 10\,000$) from the joint posterior of the fitted parameters; bands reflect parameter uncertainty only, not measurement noise at the extrapolation temperatures. (b) Residuals with respect to the digitized Liu et al. values; the ± 5 MHz gray band is a visual guide. Root-mean-square deviations are quoted in the legend. The Liu et al. values were digitized from Fig. 2(c) of Ref. [14] and carry no independently assigned measurement uncertainty.

Table 2: Comparison with the Liu et al. parametrization up to 1000 K.

T (K)	D_{Liu} (GHz)	D_{model} (GHz)	Difference (MHz)
300	2.8680	2.8699	-1.9
400	2.8600	2.8605	-0.5
500	2.8490	2.8484	$+0.6$
600	2.8360	2.8344	$+1.6$
700	2.8210	2.8194	$+1.6$
800	2.8050	2.8035	$+1.5$
900	2.7880	2.7872	$+0.8$
1000	2.7700	2.7704	-0.4

model exhibits deviations that grow rapidly outside the fitting window. Over the full 300–1000 K range, the DTQEM extrapolation yields $R^2 = 0.998185$ and an RMSE of 1.32 MHz (computed on the digitized Liu points, which carry no independently assigned measurement uncertainty). The Liu et al. parametrization lies within the DTQEM 95% parameter-uncertainty band across the full 300–1000 K range, whereas the power-law band diverges above approximately 600 K (Fig. 2a).

4 Discussion

4.1 Physical interpretation

The fitted frequency $\omega = 711 \pm 48\,\mathrm{cm}^{-1}$ lies between the acoustic-phonon regime and the zone-center optical phonon at 1332 cm^{-1}, and is bracketed by the two effective modes reported by Cambria et al. [12]. It should therefore be interpreted as an effective energy scale representing the dominant spin–phonon coupling channel, not as a spectroscopically resolved single lattice mode.

A Debye-integral model was also tested as an alternative, replacing $n(\omega, T)$ with the integrated phonon density of states weighted by a Debye cutoff frequency ω_D. Although physically motivated, this model requires numerical integration, introduces an additional free parameter, and yields no measurable improvement in R^2 or RMSE within the calibration window. The single-mode ansatz is therefore retained on grounds of parsimony.

4.2 Identifiability and scope

The model should be interpreted as a compact effective description whose individual parameters are constrained by the present dataset but not fully uniquely identifiable without broader temperature coverage. The agreement with the Liu parametrization indicates a robust thermal trend rather than a fitting artifact. The model should be used with caution above 1000 K, where higher-order phonon–phonon interactions become significant.

The uncertainty on the effective phonon frequency, $\sigma(\omega) = 48$ cm^{-1} ($\sim$7%), could be substantially reduced by incorporating low-temperature measurements (e.g., $T < 100$ K), where the exponential sensitivity of $n(\omega, T)$ to ω is strongest. Alternatively, a joint fit to both $D(T)$ and $1/T_1(T)$ over a common temperature range would provide independent constraints on the coupling amplitude A and the effective frequency ω, improving parameter identifiability without requiring new instrumentation beyond what is standard in NV-center spectroscopy.

4.3 Implications for thermometry

The model combines a physical low-temperature limit, a controlled high-temperature asymptote, and strong extrapolation performance, making it preferable to unconstrained polynomial fits for NV thermometry applications.

4.4 Possible extension to relaxation

The same phonon-occupation framework may be extended to spin-lattice relaxation rates $1/T_1(T)$. A single-mode ansatz of the form $1/T_1(T) \propto n(\omega, T)[n(\omega, T) + 1]$ can reproduce qualitative trends observed in NV centers, but a robust quantitative test would require experimental data spanning at least 100–600 K with sub-percent precision on T_1. The extension also requires verifying whether the same effective frequency ω governs both $D(T)$ and $1/T_1(T)$, or whether a second mode must be introduced. This is left as a concrete objective for Paper II of this series.

5 Conclusion

We have presented a physically motivated single-mode occupation-number model for $D(T)$ in NV centers. The model matches the best published polynomial fit within the reported experimental uncertainties of the Ouyang dataset while providing direct physical interpretability. The extracted frequency $\omega = 711 \pm 48\,\mathrm{cm}^{-1}$ is consistent with the diamond phonon spectrum. Most importantly, when calibrated only on 298–383 K, the model reproduces the Liu et al. parametrization up to 1000 K with $R^2 = 0.998185$, whereas a power-law calibration performs poorly outside the fitting window.

Data Availability

The Ouyang et al. data used for model calibration are available in Table 1 of Ref. [10]. The Liu et al. parametrization values used for extrapolation validation were digitally extracted from Fig. 2(c) of Ref. [14] and are tabulated in the present manuscript. A preprint version of this work, along with the Python fitting code and the digitized Liu et al. benchmark values, is available at DOI: `[TO BE PROVIDED]`.

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